Identifiability of a point defect model for oxide growth on AISI 347

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ABSTRACT

Mechanistic corrosion models are routinely fitted with four to six parameters to a handful of measurements and then extrapolated across service lifetimes, yet whether the data determine those parameters is almost never tested. We identify a reduced point defect model for the duplex oxide grown on Nb-stabilized AISI 347 (X6CrNiNb18-10) in simulated boiling-water-reactor hydrothermal water (240 °C, 7 MPa, 0.4 ppm dissolved O_2) from published depth-profile data at three exposure times, by differentiable physics-informed inversion validated against an exact closed-form reference. Profile likelihood, cross-checked against a 1000-member bootstrap and a prior-relaxation test, leaves one of the five fitted parameters undetermined and a second only weakly determined, while a residual decomposition puts 95% of the fit statistic on three chromium points and 80% on one. The kinetics fit the calibration data as well as an empirical power law but extrapolate very differently: at ten years the power law predicts a 3.5 to 6.3 times thicker barrier layer than the mechanistic 535 nm (bootstrap 95% interval 447–704 nm), a prediction varying 1.19 to 1.56-fold across the scanned operating envelope. A designed exposure beyond roughly 3000 hours would discriminate the two. We further document and remedy a failure mode specific to sparse-data inversion.

1. Introduction

Nb-stabilized austenitic stainless steel AISI 347 (X6CrNiNb18-10) is used in boiling-water-reactor (BWR) internals and piping for its resistance to weld-decay sensitization, yet the evolution of its passive oxide film under reactor-representative hydrothermal water has so far had no mechanistic predictive model. Current practice for predicting oxide behavior on light-water-reactor internals relies either on operational water-chemistry criteria framed around threshold electrochemical corrosion potential (ECP) and dissolved-oxygen levels [1–4], which say nothing about oxide thickness or composition, or on data-driven machine-learning models of crack-growth rate [5], which are equally silent on the underlying oxide chemistry.

The duplex oxide that austenitic stainless steels develop in high-temperature water—a Cr-rich inner spinel layer growing by solid-state transport beneath an Fe-rich outer layer formed by precipitation—is well established for the 304/316 grades [6–12], and the point defect model (PDM) of Chao, Lin, and Macdonald [13–16], in the tradition of high-field oxidation theory [17], provides the standard mechanistic description: vacancy-mediated transport under a high interfacial field, yielding closed-form thickness-versus-time laws tied to interfacial reaction kinetics. Li et al. [18] extended this framework to supercritical water. For AISI 347 specifically, Veile et al. [19] recently provided the first detailed characterization of the duplex oxide developed in simulated BWR hydrothermal water, resolving Cr-rich barrier and Fe-rich outer layers by energy-dispersive X-ray (EDX) depth profiling at three exposure times; their analysis stops at empirical per-element power-law fits, with no shared physical parameter set connecting the layers and no assessment of how far those fits can be extrapolated.

This paper adapts the supercritical-water PDM formulation [18] to the subcritical hydrothermal-water regime and identifies its kinetics from the Veile et al. dataset with a differentiable, physics-informed inversion framework. The alloy specificity of the result lies in the data, not the model structure: the reaction network contains nothing Nb-specific, and what is delivered is the first PDM identification against the only depth-profile dataset available for this alloy and environment. Beyond the identification, the framework addresses a methodological gap that recurs across four decades of PDM parameter estimation. Whether calibrated against electrochemical impedance spectra [20–22] or against depth-profile data, four-to-six-parameter models are routinely fit to three-to-six data points, and while parameter confidence intervals and correlations are sometimes reported [23], we are not aware of a formal profile-likelihood or bootstrap identifiability characterization of a PDM fit. The distinction between “the model fits” and “the data constrain the parameters” becomes consequential exactly when the fitted model is meant to extrapolate beyond its calibration window, as here. We close this gap with tools mature in adjacent fields [24–26], standard in systems biology [27] and increasingly applied to battery models [28], built directly into the inversion rather than bolted on afterward.

At the core of the framework is a physics-informed neural solver [29, 30], a neural spectral-element method

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(NSEM). Kinetic parameters are recovered by a hard-constrained inversion in which the trajectories are generated by differentiable Runge–Kutta integration, so the governing equations are satisfied exactly by construction; a classical least-squares solver serves as an independent reference, and the two agree to within 0.6% on every parameter and to identical χ^2. The same machinery emulates the forward model with two neural backbones and, in a soft-constrained joint field-and-parameter mode, extends to the spatial transport problem.

It is worth stating at the outset what that machinery is, and is not, doing here. The reduced kinetics admit a closed-form solution, and the steady spatial problem admits both an analytic reference and a Newton solve; classical methods therefore reproduce every result in this paper, and in the spatial case do so more accurately. The neural solver is not required by this problem, and we do not claim that it is. What the application provides is the complementary thing—a real-data validation of the framework against exact references, the conditions for which are precisely the conditions under which the solver is dispensable—and, out of that validation, a transferable negative result. Two further distinctions matter. The property doing the statistical work is *differentiability* rather than the neural representation: exact parameter gradients through a vectorized solver are what make profile likelihood, a thousand-member bootstrap, prior-relaxation scans and a dense operating-envelope grid affordable, and closed-form models with automatic differentiation share that property. The neural representation itself becomes necessary only where no closed form or cheap forward solve exists—coupled potential and transport fields solved self-consistently, moving-boundary inverse problems in which fields and parameters are unknown together, or constitutive functions of unknown form inferred from profile data—and the value of a validated, failure-mode-characterized framework is realized there rather than here. Our prior work demonstrated this machinery on the PDM against synthetic benchmarked data, cataloguing four training-time failure modes [31]; the present paper is its first application to real experimental data, and in carrying it out we document and remedy a fifth failure mode specific to sparse-real-data inversion: joint field-and-parameter training can settle into an equilibrium that satisfies the data loss while leaving the governing equation only approximately satisfied, exposed only by re-solving the recovered parameters through an exact forward solve. Loss-balancing pathologies of this general kind are a recognized theme in physics-informed training [32, 33], and a contemporaneous study reports a closely related diagnosis in an unrelated domain [34]; the contribution here is its documentation, its remedy, and a self-consistency check that travels with the framework.

The remainder of the paper sets out the model and the inversion methodology, and then the results: the reduced model and its differentiable identification, the identifiability and uncertainty analysis, the physics-informed inversion failure mode and its remedy, the spatial extension and zero-parameter composition predictions, the nickel enrichment zone, and the long-time predictions and operating-envelope sensitivity. A discussion and conclusions follow.

2. Model and methods

2.1. Reduced-model derivation

The apparent-parameter system (2)–(3) summarized in Section 3.1 is derived in full in the Supplementary Information (Section S1): the interfacial reaction network, the constant-field potential distribution, the constant-volume coupling that collapses the two growth rates into an affine relationship, and the log-space-stabilized closed-form solutions. The derivation also documents known errata in the source formulations [18]. The closed-form solutions of Eqs. (2)–(3) agree with Radau numerical integration to better than 6×10^{-11} nm (SI, Fig. S1), and the affine outer-layer form agrees with the closed form of Ref. [18] to better than 1×10^{-10} nm.

2.2. Experimental data

Veile et al. [19] exposed X6CrNiNb18-10 coupons to simulated BWR hydrothermal water (240°C, 7 MPa, 0.4 ppm dissolved O_2) for 72, 168, and 480 h and measured element-resolved oxide layer thicknesses by focused-ion-beam cross-sections with EDX line scans. Table 1, given with the model ladder in Section 3.2, reproduces the digitized layer-thickness means used as the inversion target throughout this paper. The digitization was checked against their own reported empirical fits: a scan-level linear-space least-squares refit of $L = k\,t^n$ to our digitized chromium data reproduces their published coefficients to four significant figures ($k = 6.522$ versus their 6.521, $n = 0.4964$ versus 0.4964), which bounds the digitization error well below the scan-to-scan scatter. For the iron layer the reported scan-to-scan physical scatter of the discrete-crystal outer layer exceeds digitization precision, so their reported layer means are used as the authoritative data. No synthetic or simulated data enters the kinetic identification; synthetic data appears only as an explicitly labeled recovery test (Section 3.4).

2.3. Statistical inversion and identifiability analysis

All fits minimize the weighted least-squares objective

$$\Phi(\theta) = \sum_{i=1}^{3} \frac{\left[L_{\mathrm{bl}}(t_i;\theta) - \bar{L}_{\mathrm{Cr},i}\right]^2}{\sigma^2_{\mathrm{Cr},i}} + \sum_{i=1}^{3} \frac{\left[L_{\mathrm{ol}}(t_i;\theta) - \bar{L}_{\mathrm{Fe},i}\right]^2}{\sigma^2_{\mathrm{Fe},i}} + \left(\frac{\ln L_0 - \ln 2\,\mathrm{nm}}{0.60}\right)^2, \tag{1}$$

where $\bar{L}$ and σ are the layer means and standard errors of Table 1 and the final term is a Gaussian prior in log-parameter space on the initial barrier-layer thickness, centered on the 1–5 nm air-formed film on high-Cr stainless steel. It is the only prior in the objective, and it is carried because the exposure-time data give L_0 no interior optimum at all: without it the estimate runs to zero (Table 5). Earlier versions of this analysis also placed a prior on $\mathrm{PBR}_{\mathrm{eff}}$, centered on 1.05 and described as the stoichiometric value. That center does not survive a chromium mass balance (Section 4), so the prior has been removed and $\mathrm{PBR}_{\mathrm{eff}}$ is determined by the likelihood alone throughout. The classical minimization uses Levenberg–Marquardt least squares in log-parameter space (which enforces positivity by construction), warm-started and verified against grid restarts; convergence tolerances are 10^{-8}. The equivalent hard-mode physics-informed inversion (Section 2.4) minimizes the same objective with the trajectories generated by differentiable Runge–Kutta integration, and is the primary engine of this work; the classical solver serves as an independent cross-check. Because the two inversions agree to within 0.6% on every parameter and to an identical χ^2, the deterministic parameter values and curves reported throughout (Figs. 3–8) are the converged optimum common to both.

Profile-likelihood scans fix one parameter on a grid of ± 1.5 natural-log units about the optimum (61 points) and re-optimize all others under the full objective (1); the $\Delta\chi^2$ values reported are computed over the data residuals alone, so a prior-dominated parameter is exposed as flat rather than masked. The parametric bootstrap resamples each layer mean from a Gaussian with its standard error (1000 members, all refits converged; positivity is guaranteed by the log-space parametrization) and refits the full objective.

Because two of the parameters are prior-dominated, the number of data-determined parameters is between three and five, and the residual degrees of freedom for the accepted five-parameter model are correspondingly between 1 and 3 for the six data points. Goodness-of-fit statements are therefore given for both endpoints of this bracket rather than a single nominal degree-of-freedom count.

2.4. Physics-informed spectral-element solver

The physics-informed forward and inverse solves use spectral-element networks built on discrete variational-representation collocation operators, applied to the PDM family in our prior work [31]; we refer to the resulting neural spectral-element-method solver as NSEM. Each field is represented by an element network evaluated on a collocation node set, physical residuals are enforced at the nodes through a first-integral or collocation formulation, loss terms are combined by a balanced-residual dynamic reweighting aggregator, and optimization proceeds in two phases (Adam, then L-BFGS). Full architecture, loss definitions, and hyperparameters for every configuration are tabulated in the SI (Section S4).

The machinery is used in three configurations (Fig. 1): a "hard" zero-dimensional inverse, in which the kinetic parameters are the only learnable quantities and the trajectories are generated by differentiable Runge–Kutta integration, so Eqs. (2)–(3) are satisfied exactly by construction; a "soft" zero-dimensional inverse, in which field networks represent the trajectories directly and are trained jointly with the parameters against a combined data-and-physics loss (the configuration in which the failure mode of Section 3.4 is exposed); and the spatial extension of Section 3.5, where no differentiable-integrator shortcut exists and the soft-mode field representation is the only option, verified against an independent Newton solve on the same node set.

Every configuration is checked against an independent reference before its results are used. The classical baseline itself is established to machine precision (Section 2.1). Two neural backbones—a multilayer perceptron (MLP) and a Kolmogorov–Arnold network (KAN) [35]—are evaluated in the same harness to test backbone sensitivity, both to the same acceptance criterion (relative L^∞ error $\leq 0.5\%$ against the closed form). All results are backed by an automated test suite (25 tests) checking parity between closed-form, Newton, and neural solutions; the suite ships with the code release (see Data and code availability).

3. Results

3.1. The reduced point defect model

The governing kinetics follow the PDM in the formulation Li et al. [18] developed for supercritical water, adapted here to subcritical hydrothermal-water conditions. The oxide Veile et al. [19] characterize on X6CrNiNb18-10 is a duplex scale (Fig. 2): a chromium-rich, nanocrystalline spinel barrier layer growing into the metal, and discrete iron-rich magnetite crystals forming a porous outer layer, with a nickel enrichment zone at the receding metal/barrier-layer interface. The base kinetics treat the nickel zone as an inert

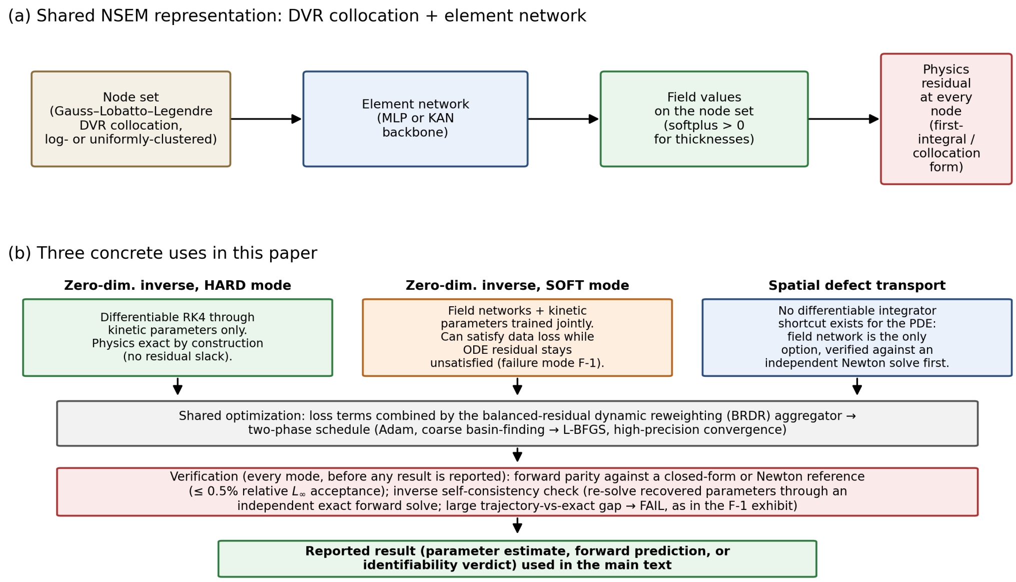

Figure 1: NSEM workflow. (a) Shared representation: field values on a discrete-variational-representation collocation node set produced by an element network (MLP or KAN backbone), with physics residuals enforced at every node. (b) The three configurations used in this work, the shared two-phase optimization, and the verification step each configuration passes before its results are reported.

consistency check; a dedicated transport model for it is developed in Section 3.6. The full reaction network, potential-distribution assumptions, and rate-law derivation are given in the Supplementary Information (SI, Section S1, which also documents known errata in the source papers), with numerical validation in Section 2.1; here we state the structure that determines the apparent-parameter system actually fit.

Two classes of interfacial reaction move the model's interfaces: barrier-layer growth into the metal, driven by oxygen-vacancy generation at the metal/barrier-layer interface, and barrier- and outer-layer destruction by chemical conversion and dissolution. Every rate constant depends on the applied potential and local pH, and the growth reaction additionally on the barrier-layer thickness through an exponential term whose sign is fixed by the PDM's constant-field assumption: growth decelerates exponentially as the film thickens, producing logarithmic-to-parabolic kinetics and, for nonzero dissolution, an eventual finite steady-state thickness. Because the open-circuit potential and pH are constant over a given exposure (the water is neutral, unbuffered, and of very low conductivity; at 240°C the neutral pH is approximately 5.6), they cannot be separated from the growth reaction's intrinsic rate constant using thickness-versus-time data alone. The potential and pH dependence is therefore absorbed into a single apparent growth prefactor, and the film-thickness dependence into a single apparent field-related exponent; it is these apparent parameters, not the underlying elementary rate constants, that the identification recovers.

The outer layer's growth is coupled to the barrier layer's through a constant-volume constraint: metal consumption at the receding metal/barrier-layer interface and barrier-layer advance proceed at the same rate, which, for negligible barrier-layer destruction in this ultrapure-water environment, collapses the two growth rates into an exact affine relationship. This reduction is algebraically simpler than the closed form of Ref. [18] and the two forms agree to better than 1×10^{-10} nm. The resulting apparent system is

$$\frac{\mathrm{d}L_{\mathrm{bl}}}{\mathrm{d}t} = A_{\mathrm{bl}}\, e^{\,b_3 L_{\mathrm{bl}}} - C_{\mathrm{bl}}, \tag{2}$$

$$\frac{\mathrm{d}L_{\mathrm{ol}}}{\mathrm{d}t} = \mathrm{PBR}_{\mathrm{eff}}\, \frac{\mathrm{d}L_{\mathrm{bl}}}{\mathrm{d}t} - C_x, \tag{3}$$

where L_{bl} and L_{ol} are the barrier- and outer-layer thicknesses, A_{bl} ($\mathrm{nm\,h^{-1}}$) is the apparent growth prefactor, $b_3 < 0$ ($\mathrm{nm^{-1}}$) the apparent field-related exponent, $C_{\mathrm{bl}} \geq 0$ ($\mathrm{nm\,h^{-1}}$) the barrier-layer dissolution rate, $\mathrm{PBR}_{\mathrm{eff}}$ the apparent (Pilling–Bedworth-type) outer-to-barrier growth-rate ratio, and C_x ($\mathrm{nm\,h^{-1}}$) a small outer-layer dissolution term; the initial conditions $L_{\mathrm{bl}}(0) = L_0$ and $L_{\mathrm{ol}}(0) = L_{\mathrm{ol},0}$ complete the seven-parameter set. The system has closed-form solutions, stabilized in log space so they remain numerically

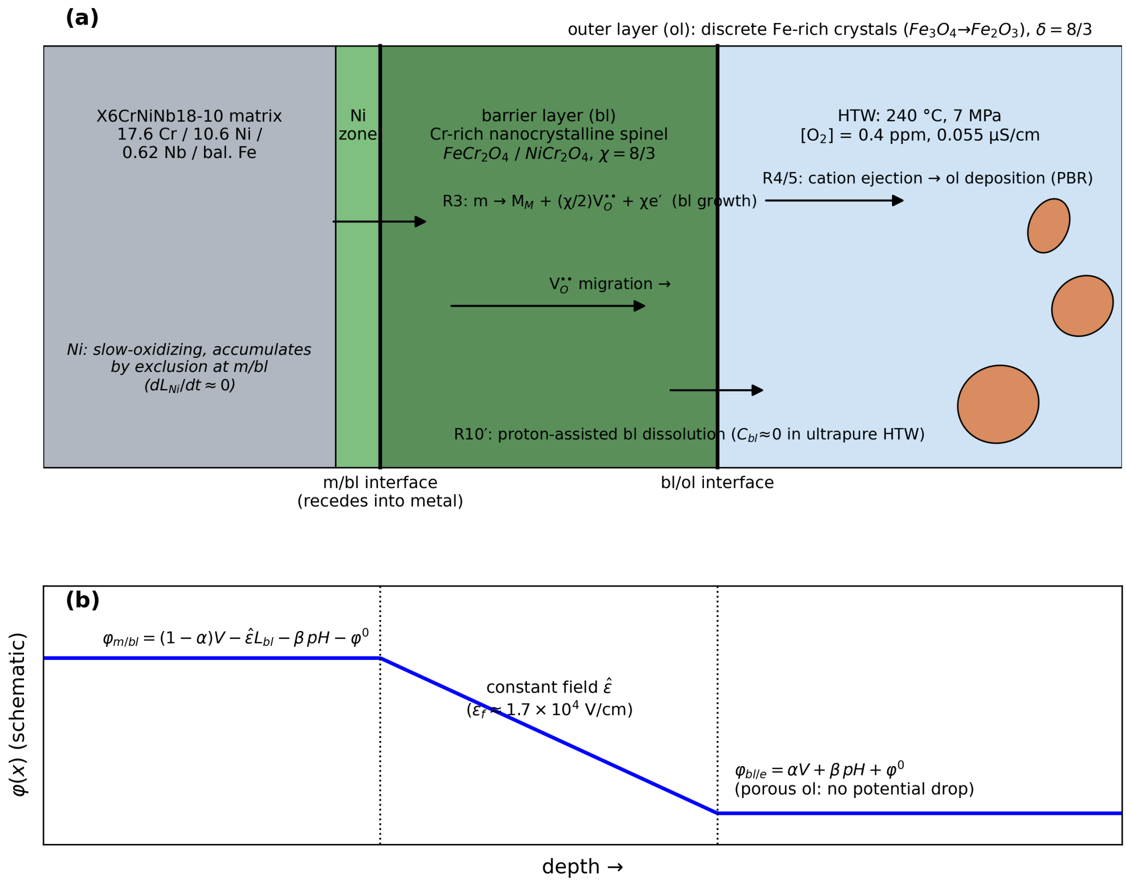

Figure 2: Duplex-oxide model schematic. (a) Layer structure (metal | Ni enrichment zone | Cr-rich nanocrystalline spinel barrier layer | Fe-rich outer-layer crystals | hydrothermal water) and the retained interfacial reactions. (b) Potential distribution across the duplex oxide under the constant-field assumption. Reaction labels and potential-drop symbols follow the SI derivation.

exact in both time limits (SI, Section S1.6). The field-related exponent maps to a field strength through $b_3 = -\alpha_3\chi\gamma\varepsilon_f$, with α_3 the growth-reaction transfer coefficient, $\chi = 8/3$ the spinel-averaged cation charge, and $\gamma = F/RT = 22.6\ \mathrm{V}^{-1}$ at 240°C; the sensitivity of the derived ε_f to the assumed α_3 is addressed in Section 4.

3.2. Differentiable inversion and the model ladder

We identify the reduced kinetics by minimizing a weighted least-squares objective (Section 2.3, Eq. (1)), solving the inverse problem two independent ways: a hard-constrained physics-informed neural inversion, in which the trajectories are produced by differentiable Runge–Kutta integration so that Eqs. (2)–(3) hold exactly, and a classical Levenberg–Marquardt solver. The two agree to within 0.6% on every parameter and to identical χ^2 (Section 3.4); we therefore report a single identified parameter set and use the ladder below only to decide model structure.

We fit the reduced system (2)–(3) to the joint Cr and Fe thickness means (Table 1) in a nested model ladder (Table 2), each variant adding degrees of freedom to test whether the flexibility is supported by the data. The four-parameter core model (M1, $C_{\mathrm{bl}} = C_x = 0$, $L_{\mathrm{ol},0} = 0$) reaches $\chi^2 = 7.15$ only by driving $\mathrm{PBR}_{\mathrm{eff}}$ to 2.43, and still cannot reproduce the disproportionately thick early iron layer under strict constant-volume coupling from zero initial thickness; adding a free dissolution rate (M2) does not help, because the fit drives that rate to zero. Releasing the coupling entirely in the six-parameter M3 variant does not produce a better model but a degenerate one, and with no prior to hold it back the degeneracy is now the interior optimum rather than something a profile has to expose: the growth prefactor and dissolution rate diverge together to $\sim 10^3\ \mathrm{nm\,h^{-1}}$ while the field parameter collapses by four orders of magnitude to $-4.4 \times 10^{-6}\ \mathrm{nm}^{-1}$, buying the lowest data χ^2 in the ladder (4.11) through an unphysical linear-growth escape route. Its ten-year barrier-layer prediction is 155 nm against M4's 535 nm, so the χ^2 ordering and the physical ordering disagree outright. That M3 wins on fit and loses on physics is the clearest statement in this paper of why χ^2 cannot by itself select a

Table 1
Layer-thickness data of Veile et al. [19] (their Fig. 9) used in this work: mean $\pm$ population standard deviation over n EDX line scans. The fits use standard errors $\sigma = \mathrm{SD}/\sqrt{n}$. The Ni enrichment zone is not a growing PDM layer and enters only the analysis of Section 3.6.

Layer	t (h)	mean (nm)	SD (nm)	n
Cr (barrier)	72	37.3	7.7	3
	168	98.3	21.7	4
	480	134.3	5.4	3
Fe (outer)	72	143.3	111.5	3
	168	225.5	142.7	4
	480	240.3	172.5	3
Ni (zone)	72	31.3	3.4	3
	168	43.0	11.1	4
	480	36.0	5.0	3

model on six data points. Parameter correlations of this kind have long been discussed qualitatively in PDM fitting; the explicit demonstration of the full ridge, its escape direction, and its capture of the unconstrained optimum on real PDM data is, to our knowledge, new.

The trap is resolved by a single additional parameter, the initial outer-layer thickness (M4): a nonzero starting population of Fe-rich outer-layer crystals captures the rapid initial precipitation Veile et al. observe by 72 h. This five-parameter model reduces the data χ^2 from 7.15 (M1) to 5.79, and does so while returning $\mathrm{PBR}_{\mathrm{eff}}$ to order unity, where M1 needed 2.43 to compensate for the missing initial precipitate population. The six-parameter M5, which adds a free dissolution rate on top of M4, reaches the same χ^2 with that rate indistinguishable from zero; it is carried forward only as the alternative long-time branch it implies (finite saturation instead of unbounded logarithmic growth, Section 3.7), not as a competing fit. For M3 and M5 the free-parameter count equals the six data points and the residual degrees of freedom are zero; their χ^2 is nonetheless nonzero because the objective (1) carries one prior residual in addition to the six data residuals, so the fit is not free to interpolate. No goodness-of-fit statistic is quoted for those rows.

The comparison with the empirical power law is reported both ways. Evaluated at Veile et al.'s published per-layer coefficients the power law gives $\chi^2 = 20.10$ against the joint layer means, but those coefficients were fit to individual scans in unweighted linear space; refit under the weighted objective (1) it reaches $\chi^2 = 7.84$ with four free parameters. On information criteria the two descriptions are then statistically indistinguishable on this dataset (Akaike criterion 15.8 for both; Bayesian criterion 14.8 mechanistic versus 15.0 power law; the small-sample-corrected criterion is undefined for M4 at $n = 6$, $n-k-1 = 0$, and is 55.8 for the power law). These criteria are quoted as indicative only at this sample size. The mechanistic model's advantage on the

Table 2
Model ladder: nested apparent-parameter models fit to the joint Cr+Fe layer-thickness data. The M3 row is the interior optimum reached from the standard starting point, and it sits on the degenerate ridge itself: the growth prefactor and dissolution rate diverge together to $\sim 10^3$ nm h^{-1} while the field parameter collapses to -4.4×10^{-6} nm^{-1}. It attains the lowest data χ^2 in the ladder and the least defensible physics, predicting 155 nm at ten years against M4's 535 nm; it is listed for completeness and rejected on structure, not on fit. The power law appears twice: evaluated at Veile et al.'s published scan-level coefficients, and refit under the same weighted objective (1) as the mechanistic models. For M3 and M5, n_{free} equals the number of data points and dof $= 0$. ‡The published-coefficient power law is *evaluated*, not fitted, against these six means—its coefficients were determined elsewhere, on individual scans in unweighted linear space—so it costs no degrees of freedom here and its residual dof is 6. No information criterion is computed for that row.

Model	Extra params	n_{free}	χ^2_{data}	dof	A_{bl}
M1 core	—	4	7.15	2	0
M2	$+C_{\mathrm{bl}}$	5	7.15	1	0
M3 (degenerate)	$+C_{\mathrm{bl}}, +C_x$	6	4.11	0	9
M4 (accepted)	$+L_{\mathrm{ol},0}$	5	**5.79**	1	**0**
M5	$+L_{\mathrm{ol},0}, +C_{\mathrm{bl}}$	6	5.79	0	0
Power law (published coeffs)	k, n per layer	0‡	20.10	6‡	
Power law (weighted refit)	k, n per layer	4	7.84	2	

calibration window is not fit quality—no comparison on three exposure times could demonstrate that—but the physical coupling it enforces: the constant-volume constraint ties the two layers through a single growth-and-precipitation picture, where the power law treats each layer as an independent curve (Fig. 3); it is this coupling that drives the divergent extrapolations of Section 3.7.

On absolute goodness of fit, $\chi^2 = 5.79$ corresponds to $p = 0.016$ at one residual degree of freedom and $p \approx 0.12$ at three (the effective-dof bracket of Section 2.3). If the mild misfit is real, the reported standard errors of the layer means understate the scan-to-scan variability—consistent with the physical heterogeneity Veile et al. report—and an error-scale correction of $s = \sqrt{\chi^2/\mathrm{dof}} = 1.4$–$2.4$ would widen every uncertainty interval below by that factor. Unscaled intervals are quoted throughout; no identifiability verdict changes under the maximal scaling (the sharply identifiable parameters remain so at 2.4× wider intervals, and the unidentifiable ones cannot get worse), but the implications for the extrapolation bands are stated explicitly in Section 3.7.

The recovered kinetics place the apparent growth prefactor at $A_{\mathrm{bl}} = 0.727$ nm/h and the field exponent at $b_3 = -0.0125$ nm^{-1} (-1.25×10^5 cm^{-1}). The recovered $\mathrm{PBR}_{\mathrm{eff}} = 1.096$ carries no prior and is therefore a statement of the likelihood alone. It sits near unity, and roughly a factor of two below the value a chromium

Table 3
Residual and χ^2 leverage decomposition of the accepted M4 fit. σ is the standard error of the layer mean (Table 1). The chromium layer carries 95% of χ^2 and the 168 h chromium point alone carries 80%.

Layer	t (h)	σ (nm)	Model (nm)	Residual (σ)	% of χ^2
Cr (barrier)	72	4.45	41.4	+0.92	14.7
	168	10.82	74.9	−2.15	80.1
	480	3.14	134.8	+0.15	0.4
Fe (outer)	72	64.37	158.9	+0.24	1.0
	168	71.35	195.6	−0.42	3.0
	480	99.59	261.2	+0.21	0.8
Total $\chi^2 = 5.79$				Cr 5.51	95.2
				Fe 0.28	4.8

mass balance on this alloy implies for an $FeCr_2O_4$ barrier. That comparison does not survive scrutiny in either direction—the mass-balance figure is phase-dependent, and the fitted parameter is not the same quantity once outer-layer loss is admitted—and is taken up in Section 4.

Where the fit statistic comes from

A five-parameter model fitted to six means invites the question of which data points actually carry the fit. Decomposing χ^2 at the M4 optimum answers it (Fig. 3e,f; Table 3). The three chromium points contribute 95% of $\chi^2 = 5.79$ and the three iron points only 5%: the outer layer's scan-to-scan scatter is so large (standard errors of the mean of 32–45% of the mean, against 2–12% for chromium) that it constrains the fit only weakly. Within the chromium set the misfit is concentrated in a single datum—the 168 h point, at -2.15σ, supplies 80% of the total—and the residual signs run +, −, + across the three exposures, a systematic pattern rather than scatter, indicating that the *curvature* of the fitted logarithmic law does not track the three measured chromium thicknesses.

Two consequences follow, and both are carried through the rest of the paper. First, the curvature parameter b_3 is what governs the long-time extrapolation of Section 3.7, so a systematic curvature residual on the only three points that constrain it is the dominant structural caveat on the ten-year number—larger, in our view, than the parametric uncertainty the bootstrap propagates. Second, the constant-volume coupling and the initial-precipitate parameter $L_{\mathrm{ol},0}$, although motivated by the iron layer, are in practice constrained by it only weakly; the mechanistic reading of the iron exponent in Section 4 should be weighed accordingly. Neither observation is grounds for preferring the power law, whose own free parameters are fitted against the same six means with the same leverage imbalance, but both bound how much any description calibrated on this dataset can claim.

3.3. Parameter identifiability and uncertainty quantification

Profile-likelihood scans referenced to the joint global minimum (Fig. 4, Table 4) show that the three exposure times sharply constrain three of the five M4 parameters: the growth prefactor, field exponent, and initial outer-layer thickness (maximum $\Delta\chi^2$ of 50, 130, and 58 across their scans). The Pilling–Bedworth ratio is only weakly identifiable (maximum $\Delta\chi^2$ 19.5, interval spanning an order of magnitude), and the initial barrier-layer thickness is formally unidentifiable from the exposure-time data (maximum $\Delta\chi^2$ of 1.0; its quoted interval simply reflects the ±1.5 natural-log scan range: the parameter is unbounded within the scanned range and is set by the air-film prior). A shallow secondary structure in the growth-prefactor profile at low parameter values is a local optimum of the compensating parameters; it sits more than six $\Delta\chi^2$ units above the global minimum and does not affect the quoted interval. Separately, an explicit search for a dissolution-rate upper bound from the thickness data finds no bound below 1 nm/h: the model's two qualitatively different long-time behaviors remain numerically indistinguishable within the exposure window, a point taken up in Section 3.7.

For a parameter carrying a prior, the profile alone cannot separate data leverage from prior leverage, because both are centred where the prior is. The direct test is to move the prior and see whether the estimate follows (Table 5). L_0 follows it completely: widening its prior fivefold moves the estimate from 1.92 to 1.14 nm (41%), and removing it drives L_0 to zero, the likelihood having no interior optimum at all. $\mathrm{PBR_{eff}}$ behaves in the opposite way. It carries no prior here, so the test is run in reverse: imposing the prior this analysis previously used, centred on 1.05 with $\sigma_{\ln} = 0.20$, moves the estimate only from 1.096 to 1.051 (4.1%) and leaves the data χ^2 at 5.79. The likelihood therefore does define an interior optimum for this parameter, and the former prior was displacing it by about 4% rather than creating it. This is the test that licenses reading $\mathrm{PBR_{eff}} \approx 1.10$ as a weak but genuine data statement while reading L_0 as prior-set.

The 1000-member parametric bootstrap cross-validates these conclusions by an independent method (Fig. 5), and the two weakly constrained parameters fail it in opposite directions. For the sharply identifiable growth prefactor and field exponent the bootstrap and profile intervals agree closely. For L_0 the bootstrap spread (±0.036 nm) is far narrower than the profile interval; this is not a tighter constraint but an artifact of the bootstrap perturbing only the data while the profile explores the full parameter range—when the data have no leverage, every resampled refit collapses onto the same prior. $\mathrm{PBR_{eff}}$, now unconstrained by any prior, shows the opposite signature: a bootstrap standard deviation of 0.93 on a central value of 1.10, with the estimate collapsing toward zero in a sizeable minority of

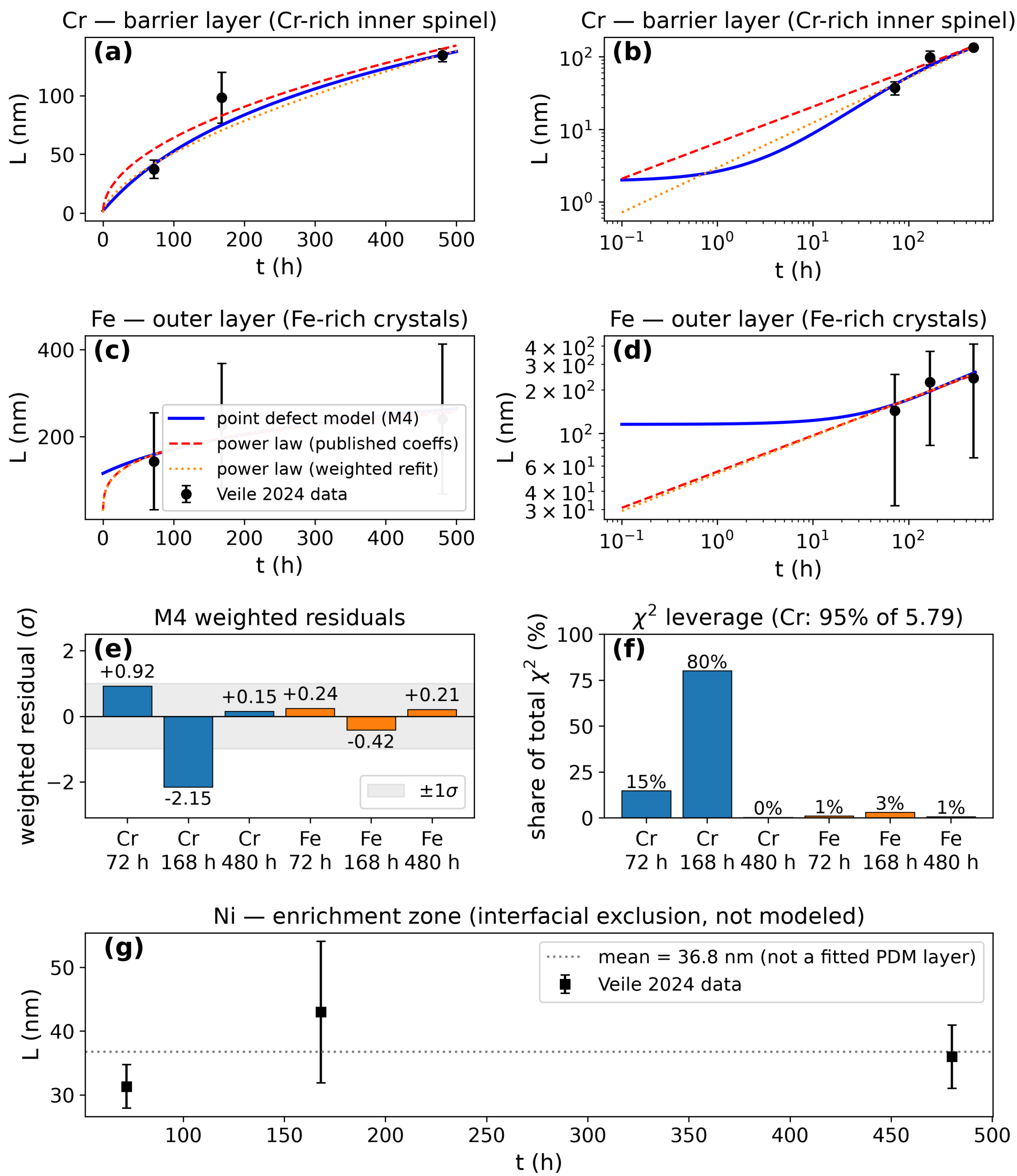

Figure 3: M4 fit and the empirical power law against the measured layer-thickness data. (a, b) Cr barrier layer, linear and log–log axes; (c, d) Fe outer layer, linear and log–log axes. Points with error bars are the means $\pm$ standard deviations of Table 1 (the fits weight residuals by standard errors, $\mathrm{SD}/\sqrt{n}$); curves are the M4 mechanistic fit and the two power-law variants (published coefficients and weighted refit). (e) Weighted residuals of the M4 fit at the six data points, and (f) each point's share of the total χ^2: the chromium layer carries 95% of the fit statistic and the 168 h chromium point alone 80%, with the $+$, $-$, $+$ chromium residual pattern indicating a systematic curvature miss rather than scatter (Table 3). (g) Ni enrichment-zone thickness, shown for completeness with its measured mean (dotted); as in Veile et al.'s own analysis, this zone is not a growing PDM layer, and its dedicated transport fit appears in Section 3.6.

Table 4

M4 parameters: fitted value, profile-likelihood 1σ interval, 1000-member bootstrap mean $\pm$ standard deviation, and identifiability verdict. The derived field strength is $\varepsilon_f = 1.7\times10^4$ V/cm at the reference transfer coefficient $\alpha_3 = 0.12$; its dominant (systematic) uncertainty is the assumed α_3, discussed in Section 4. †For L_0 the bracket gives the scan-range endpoints, not $\Delta\chi^2 = 1$ crossings: the profile never exceeds $\Delta\chi^2 = 1$ anywhere in the scanned range. No parameter carries a prior except L_0; $\mathrm{PBR_{eff}}$ is determined by the likelihood alone.

Parameter	Unit	M4 fit	Profile 1σ	Bootstrap	Verdict
A_{bl}	nm/h	0.7269	[0.626, 0.845]	0.735 ± 0.122	identifiable
b_3	nm^{-1}	-0.01249	$[-0.0145, -0.0108]$	-0.01243 ± 0.00202	identifiable
$\mathrm{PBR_{eff}}$	—	1.096	[0.245, 2.32]	1.169 ± 0.931	weakly identifiable
L_0	nm	1.9237	[0.429, 8.20]†	1.923 ± 0.0358	not identifiable (prior-set)
$L_{\mathrm{ol},0}$	nm	115.58	[25.79, 210.6]	110.5 ± 74.7	identifiable

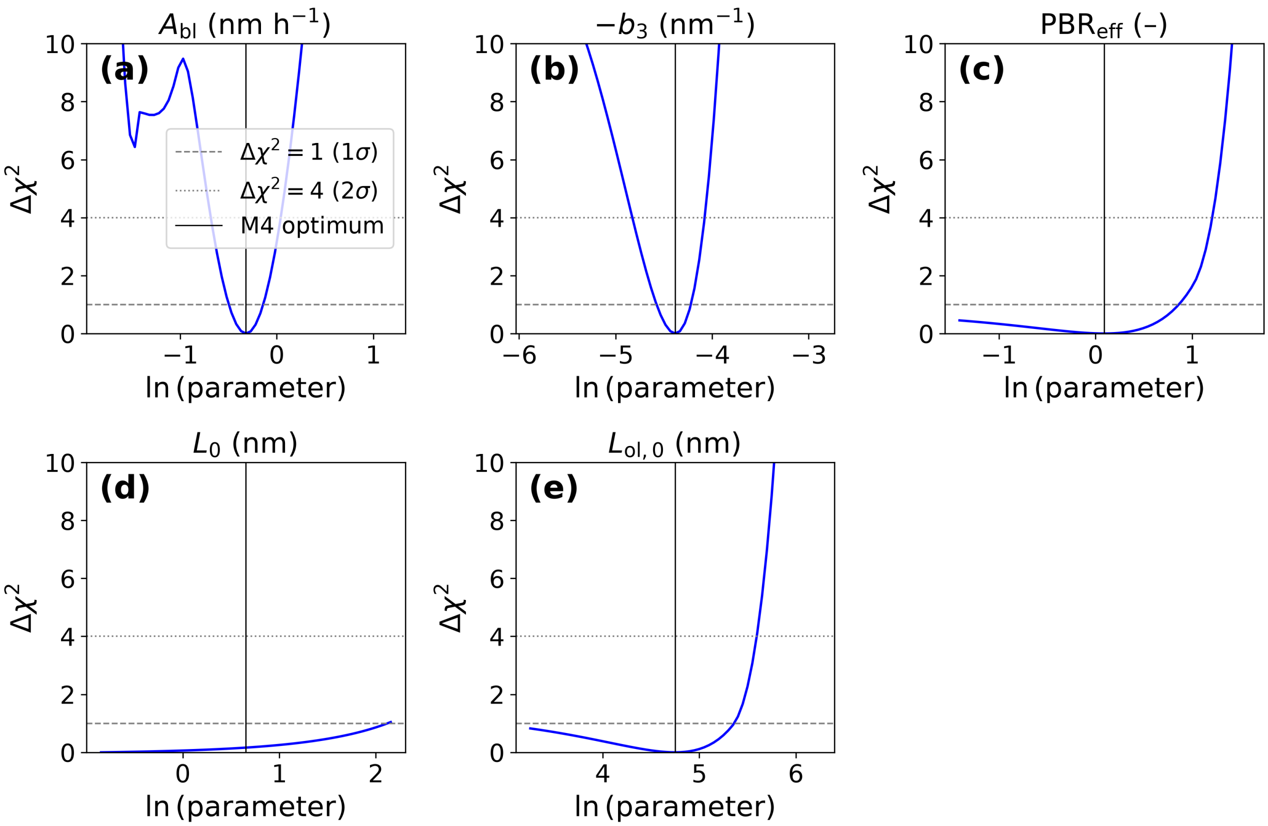

Figure 4: Profile-likelihood identifiability map: $\Delta\chi^2$ over the data residuals versus each M4 parameter (natural-log abscissa), with the 1σ ($\Delta\chi^2 = 1$) and 2σ ($\Delta\chi^2 = 4$) thresholds marked.

replicates (16th percentile 5×10^{-9}). Its profile interval [0.245, 2.32] and its bootstrap spread agree that three exposure times pin this parameter only to within about an order of magnitude. The initial outer-layer thickness sits between these cases: its profile is sharp ($\Delta\chi^2 = 58$) but its bootstrap is not (110 ± 75 nm, 16th percentile 17 nm), because $L_{\mathrm{ol},0}$ and $\mathrm{PBR_{eff}}$ trade off against each other in the outer-layer data and a resampled replicate can move along that trade-off at little cost in χ^2. The profile-likelihood interval, not the bootstrap spread, is the honest uncertainty statement for L_0; for $\mathrm{PBR_{eff}}$ the two methods agree that the constraint is weak; and for $L_{\mathrm{ol},0}$ the wider bootstrap should be preferred, since it is the one that samples the trade-off. All three verdicts are consequences of the same shortage of exposure times.

3.4. Validating the neural inversion, and a failure mode specific to sparse data

Because the reduced kinetics have a closed-form solution, this problem admits something that most physics-informed inverse problems do not: an exact reference against which the neural inversion can be graded rather than merely trusted. This section uses that reference twice—once to establish that the hard-mode inversion is interchangeable with a classical fitter,

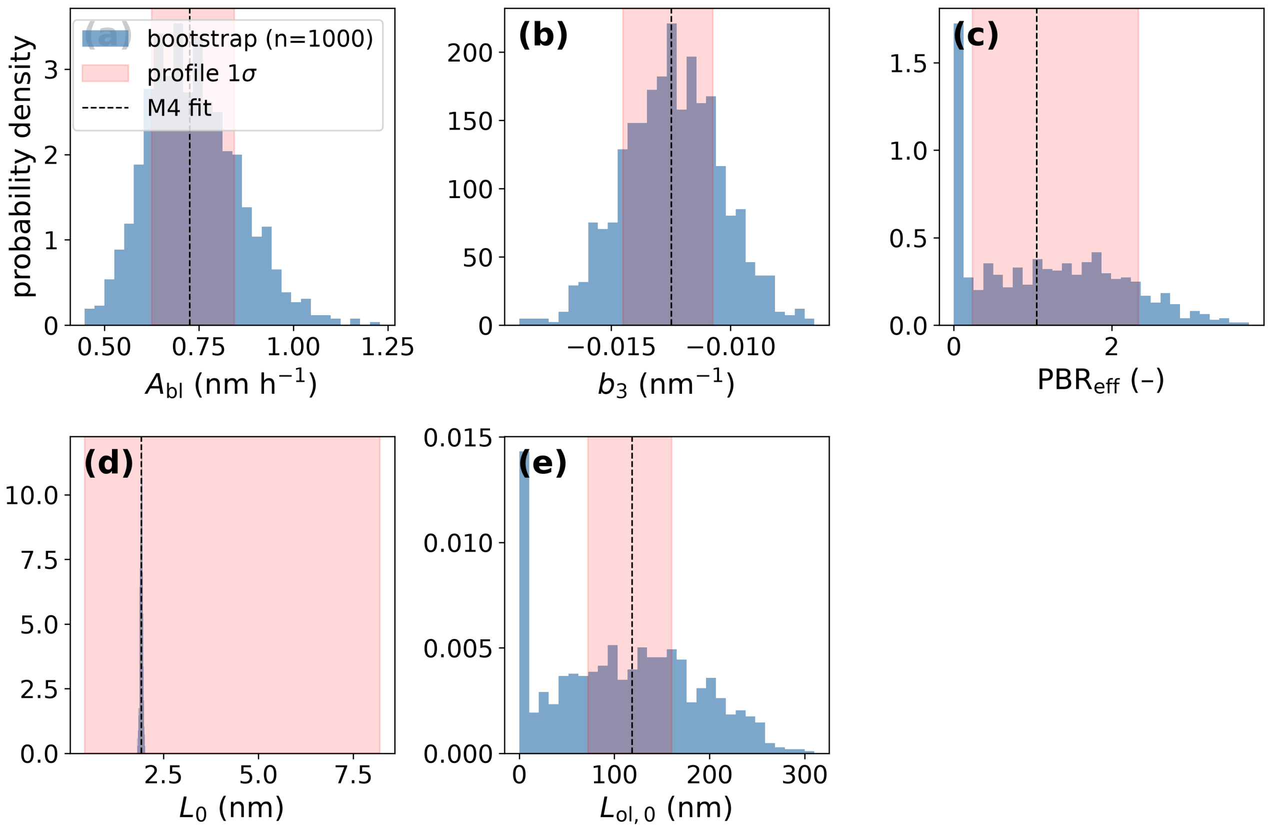

Figure 5: Bootstrap ensemble versus profile likelihood: 1000-member parametric bootstrap distribution for each M4 parameter, compared against the profile-likelihood 1σ band.

Table 5
Prior-leverage test. Each row refits M4 with the Gaussian log-space prior widths of Eq. (1) modified as shown ("off" = prior removed). A data-determined parameter does not move when its prior is relaxed; a prior-set one drifts or loses its optimum.

Prior setting	$\sigma_{\ln \mathrm{PBR}}$	$\sigma_{\ln L_0}$	$\mathrm{PBR}_{\mathrm{eff}}$	L_0 (nm)	χ^2_{data}
As fitted (baseline)	—	0.60	1.096	1.924	5.79
Impose PBR 1.05, $\sigma = 0.20$	0.20	0.60	1.051	1.924	5.79
Impose PBR 1.05, $\sigma = 1.00$	1.00	0.60	1.069	1.924	5.79
L_0 prior $\times 5$	—	3.00	1.097	1.144	5.71
L_0 prior off	—	—	1.098	0.000	5.58
Both (impose PBR, L_0 off)	0.20	—	1.051	0.000	5.59

and once to expose a soft-mode failure that the training loss does not reveal.

Applied to the real dataset, the hard-mode neural inversion recovers the deterministic M4 optimum to within 0.6% on every parameter, and its trajectory χ^2 of 5.79 matches the classical baseline exactly (Table 6). Its recovery behavior was first validated on synthetic data generated from known ground-truth kinetics with 2% relative noise at the same exposure times: recovery deviations are 1.37% (growth prefactor), 3.39% (field exponent), 0.59% (Pilling–Bedworth ratio), 11.06% (initial barrier-layer thickness), and 0.16% (initial outer-layer thickness). The one large deviation lands on exactly the parameter the profile-likelihood analysis flags as unidentifiable—a direct cross-validation of the identifiability finding by an independent method. A 50-member subset of the bootstrap re-solved through the hard-mode inversion gives spreads statistically consistent with the 1000-member deterministic ensemble (growth prefactor 0.74 ± 0.12 deterministic, 0.76 ± 0.14 physics-informed), confirming the two fitters are interchangeable.

The soft-mode inversion behaves differently. Trained on the same data, the joint field-and-parameter optimization settles into an equilibrium at which the data loss is essentially zero while the governing equation along the recovered trajectory is not satisfied to the precision the low data loss suggests. Re-solving the recovered parameters through the exact forward solution exposes the inconsistency directly: the trajectory's own χ^2 is 0.21, but the same parameters evaluated through the exact solution give $\chi^2 = 16.6$ (Table 6)—a factor of roughly 80 apart. Increasing the fixed weight on the physics residual does not resolve this, because the adaptive loss aggregator renormalizes relative term weights during training; an independent reweighting toward the data term reproduces the same signature

Table 6
F-1 self-consistency exhibit: trajectory χ^2 versus closed-form χ^2 at the recovered parameters (seed-0 runs; see SI Section S4.4 for run-to-run variability).

Inverse mode	Note	χ^2 traj.	χ^2 closed-form	Self-consistent
Hard (real data)	physics exact by construction	5.795	5.79	yes
Soft, $\lambda_{\mathsf{phys}} = 10^3$	joint field+parameter	0.21	16.64	no
Soft, $\lambda_{\mathsf{data}} = 10$	joint field+parameter	0.67	15.57	no

(trajectory 0.67 versus closed-form 15.6). The specific values vary from training run to run, but the one-to-two-orders-of-magnitude signature is stable (SI, Section S4.4). We term this failure mode F-1 and attribute it to the gradient geometry near the data manifold: once the data loss is near zero, the physics-residual gradient is too small to pull the trajectory off that manifold. This mechanism, and the expectation that dense data would prevent the imbalance, are offered as a working explanation consistent with the observations rather than as a demonstrated result.

The practical fix is the hard-mode construction, available for the zero-dimensional kinetics but not, in general, for the spatial extension. For soft-mode inversions we therefore apply, and propose for reporting alongside any physics-informed inverse result, a self-consistency check: re-solve the recovered parameters through an independent physics-exact forward solve (closed form, Runge–Kutta, or Newton iteration) and report the resulting χ^2 next to the trajectory χ^2. In our experience this is the only test among those we tried that reliably distinguishes a trustworthy soft-mode fit from an F-1 failure without requiring a hard-mode alternative to exist; its acceptance rule is a stated heuristic (SI, Section S4.4).

Finally, both neural backbones pass the 0.5% forward acceptance criterion under identical settings (relative L^∞ errors 0.14% and 0.12% for the two thickness fields with the multilayer-perceptron backbone, 0.11% and 0.04% with the Kolmogorov–Arnold-network backbone). This is a parity result: on this problem, backbone choice does not materially change forward accuracy.

3.5. Spatial extension: defect transport and composition

Veile et al.'s line scans resolve full composition-versus-depth profiles that the zero-dimensional fit never uses. We therefore extend the model spatially—resolving oxygen- and cation-vacancy transport across the barrier layer under the already-identified kinetics—and use the resulting composition predictions as an independent cross-check against data the fit never saw. The transport problem admits no differentiable-integrator shortcut of the kind the zero-dimensional kinetics allow, so the spatial solves use the soft-mode field representation; it is verified against an independent Newton solve on the same node set, which for this steady problem is both cheaper and more accurate, so

Table 7
Synthetic identifiability gate for the spatial shape parameters: 20-replicate recovery under the instrument and heterogeneity model of SI Section S2.5. RSD is the replicate relative standard deviation of the recovered parameter; the verdict thresholds are 25% (identifiable) and 50% (weak). The L_{bl} truth value is the 480-h kinetic-fit output of Table 3 rounded to 134.0 nm.

Shape param.	Physics mapping	Truth	Unit	RSD	Verd
L_{bl}	kinetic-fit output	134.0	nm	2%	ident
w_{int}	solid-state interdiffusion	5.0	nm	17%	ident
w_{out}	outer-interface kinetics	8.0	nm	19%	ident
A_{Ni}	Ni exclusion ratio	25.0	wt%	7%	ident
w_{Ni}	Ni exclusion + transport	35.0	nm	8%	ident
$f_{\mathsf{Cr,bl}}$	interfacial flux ratios	45.0	wt%	2%	ident
g_{bl}	defect-transport gradient	1.0	wt%	47%	weak

the comparison below grades the neural solver rather than relying on it. Every kinetic parameter entering this section is frozen at its zero-dimensional value.

Before any spatial inversion, a synthetic identifiability gate asks which spatial quantities the dataset could constrain at all, using an instrument model with position jitter, compositional noise, and, critically, the scan-to-scan physical heterogeneity Veile et al.'s replicate scans exhibit (SI, Section S2.5; omitting the heterogeneity term produces a spuriously optimistic verdict for every parameter). Twenty replicate synthetic experiments (Table 7) show that the barrier-layer thickness, interfacial transition widths, nickel-zone amplitude and width, and in-layer chromium fraction are identifiable (replicate relative standard deviations 2–19%), while the in-layer defect-transport composition gradient—the one quantity the spatial extension uniquely adds—is weak (47%, with a minimum statistically detectable gradient of 0.93 absolute weight percent across the layer). The spatial model's inverse scope is restricted accordingly, and the defect-transport gradient is treated as a forward prediction, not a fitting target.

The steady-state transport problem, solved by Newton iteration on a spectral collocation grid and by an NSEM forward solve, agrees with its analytic reference to 8.9×10^{-16} (oxygen vacancies) and 2.2×10^{-13} (cation vacancies); the NSEM solve agrees with the Newton reference to 2.5×10^{-4} and 5.4×10^{-5} relative error (Fig. 6). The verification also establishes a structural

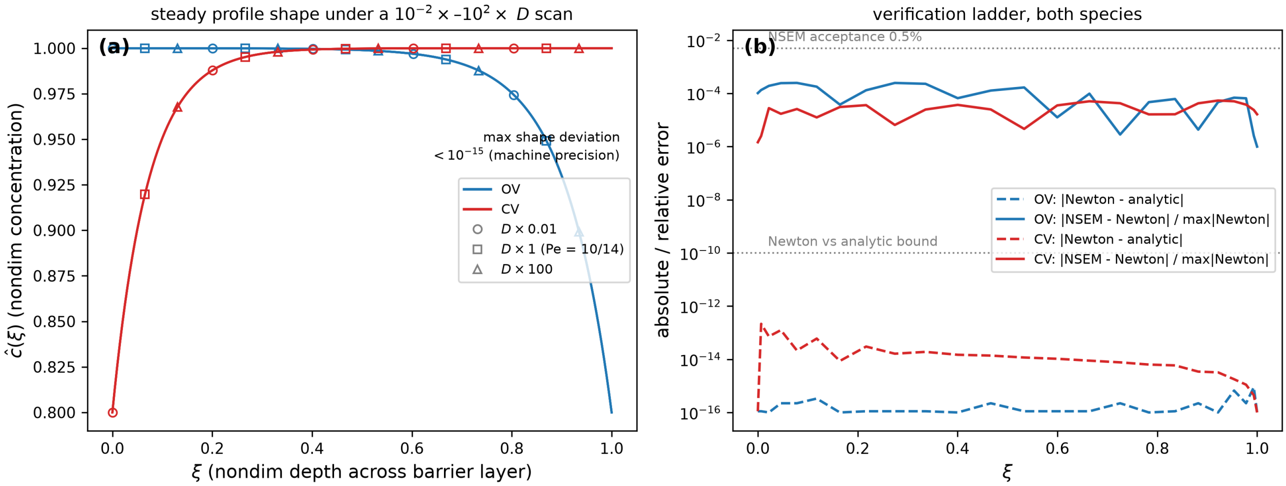

Figure 6: Steady spatial solve. (a) Steady-state nondimensional profile shape with diffusivity scaled by $0.01\times$ to $100\times$; the curves coincide to machine precision, confirming the diffusivity-cancellation result. (b) Verification ladder: Newton versus analytic, and NSEM versus Newton, for both defect species; horizontal lines mark the 10^{-10} and 0.5% acceptance bounds.

identifiability result: the steady nondimensional profile shape depends only on the Péclet-type group $\mathrm{Pe} = |z|\gamma\varepsilon_f L$, fixed by the zero-dimensional fit, and not on the defect diffusivity. This was confirmed numerically, not only by inspection of the equations: solving the identical problem with the diffusivity varied over four orders of magnitude leaves the profile shape unchanged to machine precision, while the absolute concentration scale—invisible to a composition measurement such as EDX—scales inversely with diffusivity exactly as the equations predict. Quasi-steady composition measurements therefore cannot constrain the defect diffusivity; only genuinely transient early-time composition data would carry diffusivity sensitivity, through the migration transit times quantified below. This structural argument independently corroborates the statistical gate above.

The transient extension, solved on a moving Landau-transformed domain tracking the growing layer, quantifies an approximation every zero-dimensional PDM application makes implicitly: that the defect profile equilibrates instantaneously to its steady shape. At the end of the 480-h window the fast oxygen-vacancy species (migration transit time of order 5 h) lags its quasi-steady shape by 0.69%, and the slower cation-vacancy species (transit time of order 36 h) by 5.79% (SI, Fig. S2)—both inside the acceptance tolerances used here, and both scaling with the ratio of transit time to growth timescale as the physics predicts. We are not aware of a prior explicit quantification of this quasi-steady error in the PDM literature.

The composition-map prediction uses no free parameters, and its two components test different things. The predicted chromium plateau inside the barrier layer, 46.7 weight percent against a measured plateau of approximately 45 weight percent, follows from spinel stoichiometry alone—the $FeCr_2O_4$ assignment [36] Veile et al. report as one of several candidate chromium-rich spinels (alongside Cr_2O_3, Fe_2CrO_4, and $NiCr_2O_4$) consistent with their X-ray photoelectron spectroscopy results—and therefore validates a plausible phase assignment, not the kinetics. The kinetic content is in the layer widths, which the frozen kinetics match to within $+22\%$, -20%, and $+3\%$ at the three exposure times (Fig. 7)—agreement at the tens-of-percent level from a zero-free-parameter prediction, not a quantitative reproduction—extracted from prediction and measurement by the same criterion Veile et al. use (chromium trace exceeding its far-field baseline by three weight percent). The predicted in-layer defect gradient—1.45, 0.96, and 0.46 absolute weight percent at 72, 168, and 480 h—straddles the 0.93 weight percent detection floor from the identifiability gate: marginally detectable at the earliest exposure, undetectable by the latest.

3.6. The nickel enrichment zone

The nickel zone is the one feature of the duplex scale that the identified kinetics do not describe, and the reason is instructive. An exact moving-frame nickel-transport model with a constant effective mobility (SI, Section S3), fitted to the measured zone widths and the 72-h peak amplitude, is *rejected* by the data: $\chi^2 = 25.7$ on one degree of freedom. The rejection is one of shape, not scale—the model predicts a zone width growing monotonically with exposure, while the measured widths are approximately flat (31, 43, 36 nm at 72, 168 and 480 h). Because parameter values and nested-model $\Delta\chi^2$ comparisons obtained under a structurally rejected model are not interpretable as estimates or as evidence, the fitted values and the nested-model comparison are reported in the SI

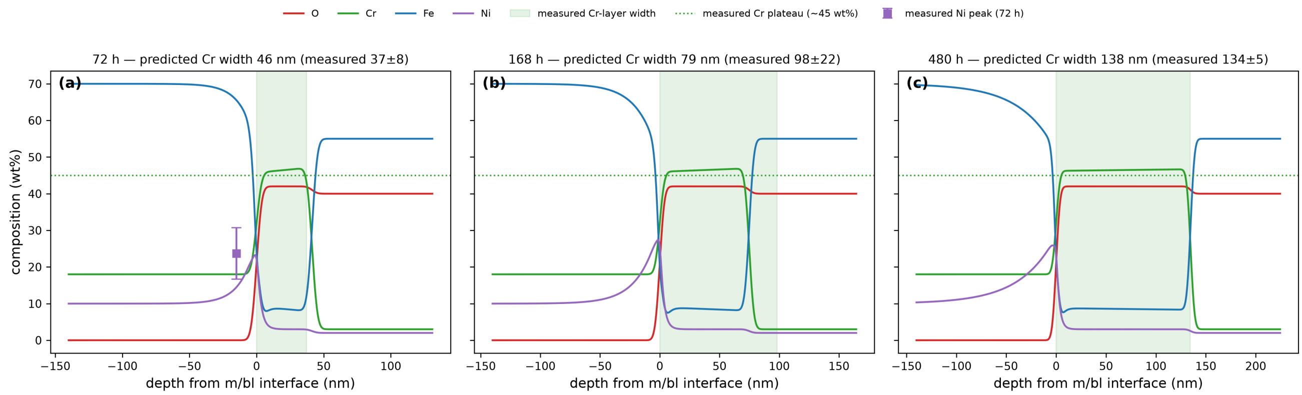

Figure 7: Zero-free-parameter composition-map prediction against Veile et al.'s measured EDX features at (a) 72, (b) 168, and (c) 480 h. Curves are predicted composition profiles (O, Cr, Fe, Ni; weight percent) versus depth from the metal/barrier-layer interface; overlays mark the measured features compared: Cr-layer width (shaded band), Cr plateau level (∼45 wt%, dotted), and the 72-h Ni-peak amplitude (23.7 ± 7.0 wt%, error bar). Full digitized EDX traces are not available, so the comparison is feature-based.

(Section S3.1, Table S6) and are not carried into the Discussion as findings.

Two statements about the nickel zone do survive independently of that fit, and they are the section's result. First, the measured widths are flat within scatter across a factor of 6.7 in exposure time, which no constant-mobility transport model of this class can produce: any such model predicts a width growing as the square root of time or faster. Second, any effective mobility consistent with the observed zone scale—a zone tens of nanometres wide developing over hundreds of hours—is of order 10^{-17} cm^2/s, which exceeds extrapolated lattice self-diffusion of nickel in austenite at 513 K by many orders of magnitude. This is a scale argument, independent of the fitted value: the zone-forming transport must be short-circuit or defect-flux mediated rather than lattice-diffusive. Taken together, the two point to a time-decaying or interface-coupled mobility.

That mechanism is exactly what three exposure times cannot resolve. Two physically motivated one-parameter extensions were tested—a defect-flux-coupled effective mobility tied to the identified growth kinetics, and a finite-capacity interfacial trapping closure—and both fail a twenty-replicate synthetic identifiability gate before touching real data (SI, Table S7): the flux-coupled closure's new parameter is recoverable but destabilizes the already-identified transport parameters past threshold, and the finite-capacity closure's parameter is not recoverable at all (SI, Section S3.2). A forward-only scan across the flux-coupled closure's plausible range confirms no single value is simultaneously shape-correct and a quantitative improvement. Resolving the nickel zone is therefore left as a target for denser exposure-time data, and is one more instance of the exposure-time limitation that recurs throughout this paper.

3.7. Long-time extrapolation and operating-envelope sensitivity

At ten years—a factor of roughly 180 beyond the 480-h calibration window—the identified model predicts a barrier-layer thickness of 535 nm, with bootstrap 68% and 95% intervals of [482, 606] and [447, 704] nm (Fig. 8). These bands quantify parametric uncertainty conditional on the M4 model structure and on the reported standard errors; under the maximal error-scale correction of Section 3.2 ($s = 2.4$, applied in log space) the 95% interval widens to approximately [340, 1040] nm. Extrapolating Veile et al.'s published power-law coefficients over the same horizon instead predicts 1853 nm (a factor of 3.5), and the weighted-refit power law 3375 nm (a factor of 6.3; bootstrap 68% interval on the ratio [5.6, 7.0]). Either power-law prediction lies outside the mechanistic band even after the maximal inflation. The divergence is the practical consequence of the constant-volume-coupled growth law saturating logarithmically while the power law, fit independently to each layer, extrapolates indefinitely.

The model's own two long-time branches—unbounded logarithmic growth (M4) versus eventual saturation (M5)—track within about five percent of each other out to 10^5 years: no feasible exposure experiment could distinguish them from thickness data, the prediction-space restatement of the dissolution-rate unidentifiability of Section 3.3. Discriminating mechanistic from power-law extrapolation is, by contrast, feasible. Under a conservative scatter model for a future campaign (the worst observed chromium scan-level relative standard deviation, 22%, averaged over three scans, giving a 12.7% standard error), the separation between the two predictions first exceeds twice that scatter at 2551 h with the published coefficients (1437 h refit) and three times at 4011 h (2024 h refit). These thresholds

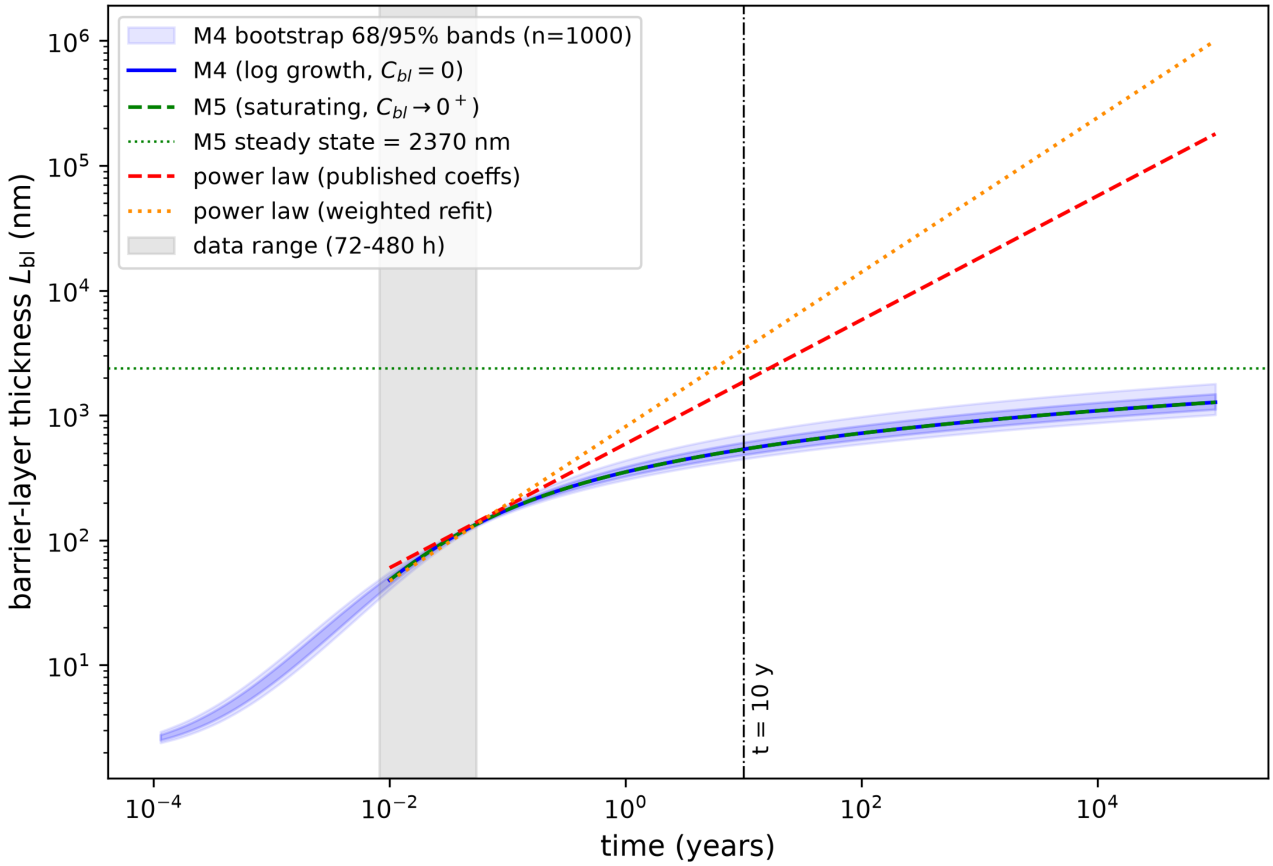

Figure 8: Long-time prediction with propagated parametric uncertainty: M4 (logarithmic growth) versus M5 (saturating branch) versus the two power-law variants, to 10^5 years. Shaded bands are the 68% and 95% intervals of the 1000-member parametric bootstrap propagated through the M4 closed form, conditional on the M4 model structure and the reported standard errors; the M5 curve is indistinguishable from M4 at all feasible exposures. The vertical line marks the ten-year horizon.

compare the separation to future measurement scatter alone, and so understate the required exposure. Adding the mechanistic prediction's own parametric uncertainty (the 1000-member bootstrap propagated to each exposure time) and the power-law fit uncertainty (an equivalent bootstrap of the weighted refit) in quadrature moves the 2σ threshold to 2996 h with the published coefficients and 1944 h with the refit, and the 3σ threshold to 5133 h and 5850 h respectively. A designed exposure of roughly 3000 h is therefore the realistic requirement for 2σ discrimination against the full error budget, and roughly 5000–6000 h for 3σ; the scatter-only numbers should be read as the optimistic bound they are.

Because the dataset was collected at a single temperature and oxygen level, the model's sensitivity is mapped across a scanned operating envelope of 200–285°C at 7 MPa and 0.01–8 ppm dissolved oxygen—deliberately extended beyond normal-water-chemistry oxygen levels (∼0.2–0.4 ppm) to bound off-normal chemistry. The upper temperature bound is the liquid-phase stability limit at 7 MPa ($T_{\mathrm{sat}} \approx 286$°C), so nominal BWR coolant temperature (≈ 288°C at operating pressure) sits essentially at the edge of the scanned box, not comfortably inside it. The temperature channel follows the rate-constant structure of the supercritical-water formulation [18]: the rate-determining constant scales as $\exp[-\alpha_3 \Delta G_R^0 / R\,(1/T - 1/T_0)]$, so the effective Arrhenius activation energy is $\alpha_3 \Delta G_R^0$, and with the standard Gibbs energy of the growth reaction unmeasurable from single-temperature data, ΔG_R^0 must be scanned explicitly. The scan is set by the effective activation energy it implies rather than by ΔG_R^0 itself: at $\alpha_3 = 0.12$ the range 25–417 kJ/mol corresponds to $\alpha_3 \Delta G_R^0 = 3$–50 kJ/mol, which spans both the low values implied by transferring a supercritical-water transfer coefficient and the several tens of kJ/mol reported for weight-gain kinetics of austenitic steels in high-temperature water. Scanning only the lower part of that range would build the robustness conclusion on the assumption most favourable to it. A field-attenuation channel is tied to the same temperature

Table 8
Operating-envelope spread of the predicted barrier-layer thickness over the scanned box (200–285°C at 7 MPa, 0.01–8 ppm O_2), as a function of the assumed effective Arrhenius activation energy $\alpha_3 \Delta G_R^0$. The first three rows are the originally scanned range; the last two extend it to the values reported for weight-gain kinetics on austenitic steels in high-temperature water.

ΔG_R^0 (kJ/mol)	$\alpha_3 \Delta G_R^0$ (kJ/mol)	L_{bl}(480 h) range (nm)	L_{bl}(480 h) factor	L_{bl}(10 y) range (nm)	L_{bl}(10 y) factor
25	3	123–147	1.20	492–583	1.19
50	6	119–151	1.27	487–588	1.21
100	12	113–159	1.42	479–598	1.25
250	30	93–185	1.99	452–627	1.39
417	50	73–214	2.92	423–660	1.56

dependence, and the dissolved-oxygen channel is an ideal-Nernst shift of the growth prefactor—an explicit lower bound on the true corrosion-potential response (Section 4).

Across this envelope (Fig. 9, Table 8), the predicted barrier-layer thickness varies by a factor of 1.20 to 2.92 at 480 h and 1.19 to 1.56 at ten years, depending on the scanned effective activation energy; temperature dominates through the combined Arrhenius and field-attenuation channels, while dissolved oxygen contributes a mild tilt. Two features of this result matter more than the headline range. First, the sensitivity is strongly conditional on the activation energy: over the low 3–12 kJ/mol sub-range the ten-year envelope factor is a nearly negligible 1.19–1.25, but at 50 kJ/mol it is 1.56, and the 480-h factor roughly doubles from 1.42 to 2.92. Second, the ten-year factor is consistently smaller than the 480-h factor, because logarithmic growth compresses prefactor differences at long times—the robustness of the *long-time* prediction is a property of the growth law, and is not inherited from a robustness of the short-time behaviour. The long-time prediction is therefore moderately, not strongly, sensitive to operating condition across the full plausible activation-energy range, and a factor of about 1.6 should be carried alongside the parametric band of Fig. 8 rather than the 1.25 the narrow scan alone would suggest.

4. Discussion

4.1. Mechanistic reading of the empirical exponents

The mechanistic model supports a physical reading of Veile et al.'s fitted exponents that the power-law description cannot offer. The near-parabolic chromium exponent ($n \approx 0.50$) is the signature of a barrier layer growing under near-zero dissolution in ultrapure water; the sub-parabolic iron exponent ($n \approx 0.22$) reflects deposition constrained by constant-volume coupling to the barrier layer from a substantial initial precipitate population, rather than independent iron-layer kinetics. This reading is available regardless of fit quality, and it is the coupling behind it—not χ^2—that separates the two descriptions' ten-year predictions by a factor of 3.5 to 6.3.

4.2. The Pilling–Bedworth ratio and a chromium mass balance

The identified $\mathrm{PBR_{eff}} = 1.096$ is one of the two quantities the exposure-time data constrain only weakly, and it is worth being explicit about what it does and does not agree with. An earlier version of this analysis placed a prior on it centred on 1.05 and described that centre as the stoichiometric spinel/magnetite value. It is not. The constant-volume closure that yields 1.05 requires the barrier layer to draw approximately 30 at% chromium from an alloy containing 18.8 at%, which the alloy cannot supply. A chromium mass balance on X6CrNiNb18-10 with an $FeCr_2O_4$ barrier gives $\mathrm{PBR_{eff}} = 2.09$ if every iron atom not retained in the nickel-enriched zone reports to the outer layer. That figure is the natural point of comparison, but—as the rest of this section shows—it is neither unique nor a bound.

The likelihood puts the parameter at 1.10, a factor of 1.9 below that figure. That gap is not a conflict, for two independent reasons, and neither of them requires the fit to be wrong.

The first is that 2.09 is not a bound but one value of a phase-dependent quantity. It follows from assigning the barrier the composition $FeCr_2O_4$, and the photoelectron spectroscopy admits three further chromium-rich spinels. Carrying the same mass balance through each of them—using lattice-parameter molar volumes, the nominal alloy composition, and, in one variant, the nickel routing fraction identified in Section 3.5 rather than an assumption of complete nickel rejection—gives production ratios of 2.05 ($FeCr_2O_4$), 2.50 ($NiCr_2O_4$), 3.64 (Cr_2O_3) and 0.52 (Fe_2CrO_4). The stoichiometrically admissible range is therefore $[0.52, 3.79]$, and the identified 1.10 lies inside it. Only the $FeCr_2O_4$ assignment produces a discrepancy at all.

The second reason is that, at any fixed phase assignment, the two numbers are not the same quantity. A mass balance fixes the rate at which barrier growth *produces* outer-layer oxide. The accepted model pins $C_x = 0$, so its $\mathrm{PBR_{eff}}$ measures net *accumulation* instead, and the two differ by whatever oxide leaves for the coolant. Freeing C_x separates them, and doing so shows the data cannot: imposing the $FeCr_2O_4$ value of 2.05 changes χ^2 from 5.79 to 5.67, a *decrease* of 0.12 (Table 9), paid for by an outer-layer loss rate of 0.24 nm/h and a smaller initial precipitate population, 94 against 116 nm. Profiling $\mathrm{PBR_{eff}}$ from 0.5 to 8 in this family moves χ^2 by 0.89 in total, so the $\Delta\chi^2 = 1$ interval is the entire scanned range. The reason is structural rather than statistical: with $L_{\mathrm{bl}}(t)$ fixed, the outer-layer solution $L_{\mathrm{ol}}(t) = L_{\mathrm{ol},0} + \mathrm{PBR_{eff}}\,(L_{\mathrm{bl}} -$

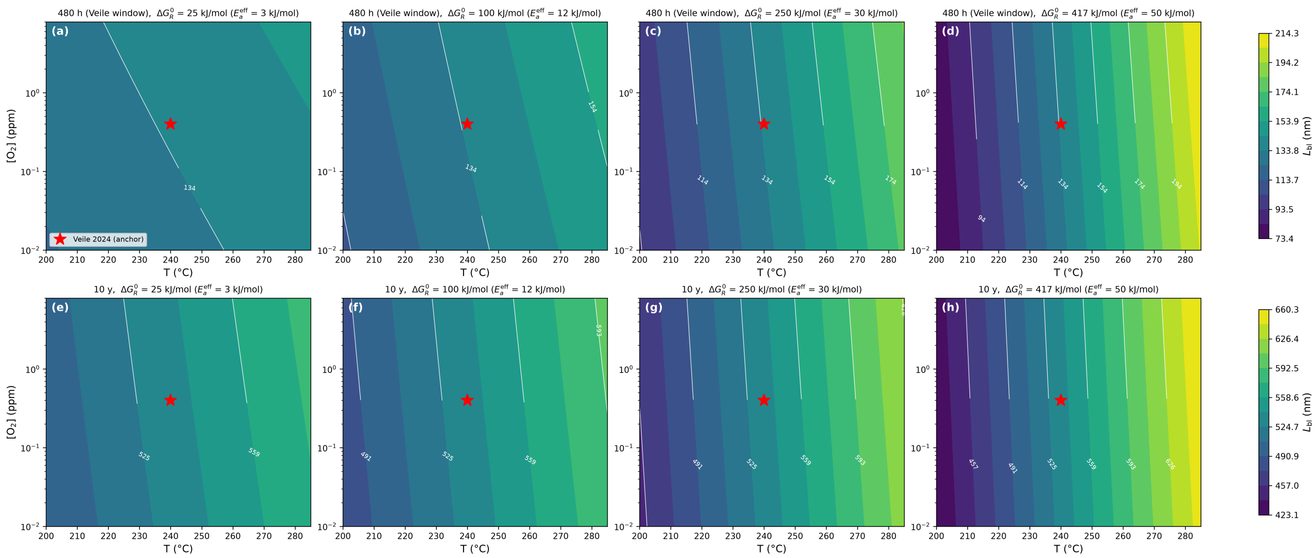

Figure 9: Operating-envelope sensitivity maps: predicted L_{bl} at (a–d) 480 h and (e–h) ten years over the scanned envelope (200–285°C, 0.01–8 ppm O_2), for $\Delta G_R^0 = 25$, 100, 250 and 417 kJ/mol—effective activation energies $\alpha_3 \Delta G_R^0$ of 3, 12, 30 and 50 kJ/mol, spanning the full plausible range rather than only the low values a transferred supercritical-water transfer coefficient implies. The marker locates the Veile et al. calibration condition. Each row shares a common color scale, so the widening spread from left to right is the sensitivity increasing with activation energy.

$L_0) + C_x t$ carries three parameters against exactly three outer-layer observations, so it is saturated and no combination is preferred.

The correct statement is therefore not that the identification disagrees with stoichiometry, but that three exposure times cannot separate oxide production from oxide loss. This also explains an observation that would otherwise look like a loose end: the M1 and M2 variants, which lack the initial precipitate population, reach $\mathrm{PBR}_{\mathrm{eff}} = 2.43$—they are simply at a different point on the same flat direction. The resolution makes one falsifiable prediction. If the barrier is $FeCr_2O_4$ and stoichiometry holds, the outer layer must lose about 9 µg of iron per square decimetre per hour, a magnitude consistent with the corrosion-product release reported for stainless steels in high-temperature water [12]. Nothing measured here demonstrates it: this dataset contains no coolant chemistry, and the outer layer's scan-to-scan scatter is so large that it carries only 5% of the fit statistic (Table 3). A coolant mass balance, or outer-layer thicknesses at more exposure times, would decide it.

4.3. What is, and is not, specific to the stabilized alloy

The reaction network fitted here contains nothing niobium-specific, and none of the five identified parameters can be attributed to Nb stabilization on the evidence presented. It is worth being explicit about what that means, since the alloy's stabilization is the reason it is used in reactor internals. The identification is specific to AISI 347 in the sense that the numbers come from measurements on AISI 347, not in the sense that the model structure encodes Nb; every parameter is an apparent quantity for an 18Cr–10Ni austenitic matrix under this environment, and the same structure fitted to 304 or 316 data in the same environment would differ only through the fitted values. Distinguishing a genuine stabilization effect would require the comparison this dataset does not contain: matched exposures of stabilized and unstabilized grades, in which NbC precipitates could be shown to alter local chromium availability at the metal/barrier-layer interface, to nucleate or pin the inner spinel, or to change the initial film that L_0 and $L_{\mathrm{ol},0}$ parametrize. Two of the identified quantities are the natural place such an effect would appear—the growth prefactor A_{bl}, through chromium supply to the growth reaction, and the initial-condition parameters, through surface and precipitate structure—and both are reported here with intervals wide enough that a moderate stabilization effect would be invisible. The claim we make is therefore

Table 9
Imposing a stoichiometric $\mathrm{PBR}_{\mathrm{eff}}$ once the outer-layer loss term C_x is free. The accepted model's value is recovered at $C_x \approx 0$; every larger value is reached at no cost in χ^2, paid for by outer-layer loss and a smaller initial precipitate population. $\chi^2 = 5.79$ for the accepted model.

$\mathrm{PBR}_{\mathrm{eff}}$ imposed	χ^2	$\Delta\chi^2$	C_x (nm/h)	$L_{\mathrm{ol},0}$ (nm
1.10 (accepted model)	5.784	−0.007	−0.025	120.2
2.05 ($FeCr_2O_4$)	5.671	−0.120	−0.241	93.6
2.16 ($FeCr_2O_4$, Ni routed)	5.661	−0.131	−0.264	90.7
2.43 (M1/M2 optimum)	5.635	−0.157	−0.327	83.0

Table 10
The identified kinetics in the context of related systems. Literature thickness entries are indicative orders of magnitude for the Cr-rich inner layer; field strengths are apparent values as reported by the cited analyses.

System	Environment	T (°C)	Inner layer	ε_f (V/cm)
AISI 347 (this work)	simulated BWR HTW	240	41–135 nm (72–480 h)	4×10^3–2×10^4
304/316 SS	oxygenated HTW [9, 11]	288	tens–hundreds nm	not reported
HCM12A, 316L	supercritical water [18]	500	μm scale	$\sim 10^2$
Passive metals	ambient aqueous [17]	25	1–5 nm	$\sim 10^6$

that this is the first mechanistic identification for this alloy in this environment, not that the mechanism identified is peculiar to the stabilized grade.

4.4. Field strength and comparison with other systems

Converting the identified field exponent with $b_3 = -\alpha_3\chi\gamma\varepsilon_f$ requires the growth-reaction transfer coefficient, which single-temperature thickness data cannot supply. At the reference value $\alpha_3 = 0.12$ fit by Li et al. [18] for a ferritic steel in supercritical water, $\varepsilon_f = 1.7 \times 10^4$ V/cm; over the physically defensible range $\alpha_3 \in [0.1, 0.5]$ the derived field spans 4.1×10^3 to 2.1×10^4 V/cm. The implied potential drop across the 480-h barrier layer, $\varepsilon_f L$, is correspondingly 0.06–0.28 V; mixed-potential estimates of the corrosion potential in oxygenated 288°C BWR water are of order 0 to +150 mV on the standard hydrogen electrode (SHE) scale at 0.2–0.4 ppm O_2 [1, 2], which favors the lower end of the derived range and is one more reason to treat ε_f here as an order-of-magnitude quantity. Within that reading, the value sits plausibly between the $\sim 10^2$ V/cm of supercritical-water PDM fits at 500°C [18] and the $\sim 10^6$ V/cm of room-temperature passive films [17]; we offer this ordering as a weak plausibility remark only, since the three anchors span different alloy classes and film chemistries, and the nearest-temperature precedents (PDM and mixed-conduction analyses of 304/316 in 280–320°C water [20, 21, 37]) do not report directly comparable apparent field strengths.

Table 10 places the identified kinetics in the context of the high-temperature-water oxide corpus. The inner-layer thickness trajectory recovered here (41 nm at 72 h to 135 nm at 480 h) is of the same order as inner-layer thicknesses reported for 304/316 stainless steels in oxygenated 288°C water at comparable times [9, 11], and the duplex morphology, inner-layer spinel chemistry, and near-parabolic inner growth match the established picture for austenitic steels in this environment [6–8]. The ten-year prediction of 535 nm has no direct validation data for AISI 347; we are not aware of published long-term harvested-component oxide thicknesses for this alloy in BWR conditions, and the prediction should be read as a testable statement, not a validated one.

4.5. The nickel zone: an anomalous mobility

The one nickel result that does not depend on a fitted model is a scale argument, and it is the mechanistically interesting one. An effective mobility of order 10^{-17} cm^2/s—the scale required for a zone tens of nanometres wide to develop over hundreds of hours at 240°C—exceeds extrapolated lattice self-diffusion of nickel in austenite at 513 K by many orders of magnitude. Whatever transports nickel at the receding metal/barrier-layer interface is therefore short-circuit or defect-flux mediated rather than lattice-diffusive. That is also the class of mechanism whose mobility would decay as the interface structure coarsens, which is the natural explanation for the flat measured zone widths that defeat the constant-mobility model (Section 3.6). We stop there: distinguishing candidate short-circuit mechanisms requires exposure-time resolution this dataset does not have, and the constant-mobility fit is rejected by the data in shape, so neither its incorporation ratio nor its nested-model comparison against a re-precipitation-only outer-layer supply picture can be offered as evidence either way.

4.6. Limitations

The kinetic identification rests on a single temperature and a single dissolved-oxygen level; the operating-envelope maps are forward projections under sourced assumptions, not validation against multi-condition data, which does not exist for this alloy. Three specific gaps between the calibration environment and an operating reactor deserve explicit statement beyond the envelope scan. First, the exposures are unirradiated and static: radiolysis products, especially H_2O_2, are known to alter oxide structure and thickness on stainless steels relative to O_2 at comparable corrosion potential [11], and flow affects outer-layer magnetite growth and dissolution through solubility and mass transfer [12]; neither channel is in the model. Second, surface finish conditions the initial film and the early precipitate population—the parameters L_0 and $L_{\mathrm{ol},0}$ here—and mechanical surface treatments are documented to change oxide properties on 304L in simulated BWR/PWR water [38]; the identified $L_{\mathrm{ol},0} = 116$ nm is specific to Veile et al.'s coupon preparation. Third, the open-circuit potential is absorbed into the fitted prefactor, so the ideal-Nernst oxygen channel of the sensitivity maps is a lower bound on the true corrosion-potential response; measured ECP–oxygen curves in BWR-relevant chemistry are steeper than Nernstian in the parts-per-billion transition region [2], and the true oxygen sensitivity may exceed what is mapped. The envelope scan now covers effective activation energies up to 50 kJ/mol (Table 8), so the temperature channel is no longer conditional on the low values a transferred supercritical-water transfer coefficient implies; what remains conditional is the channel *structure*, which assumes a single rate-determining step with a temperature-independent field strength. Finally, EDX-derived layer thicknesses are not total oxide mass, so the composition comparison is a shape and plateau-composition cross-check, not a mass-balance validation.

A further conditionality attaches to the field strength itself. Every number derived from b_3 through $b_3 = -\alpha_3\chi\gamma\varepsilon_f$ inherits the PDM's constant-field closure, and that closure can be tested directly by solving Poisson's equation together with the defect transport instead of imposing the field on it. We have carried out that solve, verifying it by reduction: with the space charge switched off it reproduces the analytic constant-field solution to 1×10^{-15}. Switched on, it does not support the closure. At the identified kinetics and the transport constants assumed in Section 3.5, the dimensionless space-charge parameter is of order 10^4–10^5 and the corresponding Debye length is approximately 0.06 nm, an order of magnitude below an interatomic spacing; holding the field within 10% of uniform would require defect diffusivities near 10^{-9}–10^{-10} $\mathrm{cm^2\,s^{-1}}$, some six orders of magnitude above the assumed values, or equivalently defect concentrations below about 10^{14} $\mathrm{cm^{-3}}$. The implication is not that the field distribution is merely non-uniform but that a two-species charged-defect description cannot produce a constant field at the concentrations the model itself implies: charge compensation by electronic carriers, which the PDM does not carry, is required. This leaves the fitted apparent parameters and every prediction drawn from them untouched, since the identification is of b_3 and not of ε_f. What it bounds is the interpretation: ε_f should be read as an apparent field-related grouping, and the numerical field strengths compared in Section 4 carry that caveat.

A limitation of a different kind attaches to the inversion machinery rather than to the chemistry. Nothing in this paper's physical conclusions requires the neural solver: the kinetics have a closed form, the steady transport problem has an analytic reference, and the classical methods used to check the neural results would have produced the same numbers—faster, and in the spatial case to eleven more digits. That is a deliberate property of the test case, not an oversight, since a validation is only possible where an exact reference exists. But it bounds the claim: what is demonstrated here is that the framework is trustworthy on real sparse data once the F-1 check is applied, not that it was needed to obtain these results. The problems that would need it are visible from this study. Two lie inside the present model—solving the potential distribution self-consistently with defect transport instead of imposing the constant-field assumption, and inverting the moving-boundary transient for kinetic parameters and defect fields together, where the classical route requires a remeshed forward solve inside every optimizer iteration. A third is sharper: the nickel zone rejects every constant-mobility description the data can test, so the quantity actually wanted there is a mobility *function* of unknown form, which a neural representation can carry as an unknown closure constrained by the transport equation while a classical solver must first commit to a parametric guess. Establishing the framework's reliability against exact references, as done here, is the prerequisite for trusting it in those settings, where no such reference is available.

Every identifiability limitation in this paper—the zero-dimensional profile analysis, the residual-leverage imbalance, the spatial gate, and the nickel-closure gate—traces to the same root cause: too few independent exposure times for the parameters genuinely in play. That convergence, and the concrete exposure-duration remedy it points to, is the paper's central methodological conclusion. More broadly, the framework—differentiable inversion with exact physics, identifiability and uncertainty quantification built in, and a self-consistency check that flags physics-informed training failures—is not specific to this alloy or to the point defect model, and applies to any sparse-data mechanistic identification where extrapolation beyond the calibration window is the purpose of the fit.

5. Conclusions

1. A differentiable, physics-informed inversion framework has identified a five-parameter reduced point defect model, adapted from the supercritical-water formulation to subcritical BWR hydrothermal water, for X6CrNiNb18-10 (AISI 347) from published depth-profile data. The hard-constrained neural inversion satisfies the governing equations exactly and agrees with a classical Levenberg–Marquardt solver to within 0.6% on every parameter, with bootstrap ensembles statistically consistent between the two fitters—the first mechanistic passive-film growth model identified for this alloy in this environment, though the model structure is not itself Nb-specific. A residual-leverage decomposition shows the three chromium points carry 95% of the fit statistic and the 168 h point alone 80%, with a systematic +, −, + residual pattern: the fitted curvature, which governs the extrapolation, is the least well determined feature of the fit.

2. Refit under the same weighted objective, the empirical power law fits the calibration data comparably (χ^2 5.79 versus 7.84), but the two descriptions extrapolate very differently: at ten years the power law predicts 3.5 to 6.3 times more barrier-layer growth than the mechanistic 535 nm (bootstrap 95% interval 447–704 nm, widening to approximately 340–1040 nm under the maximal error-scale correction); both power-law variants remain outside the mechanistic band in every case.
3. Formal identifiability analysis, cross-validated between profile likelihood and a 1000-member parametric bootstrap, shows the three exposure times sharply constrain the growth prefactor, field exponent, and initial outer-layer thickness; constrain the effective Pilling–Bedworth ratio only to within about an order of magnitude (bootstrap 1.10 ± 0.93); and cannot constrain the initial barrier-layer thickness or bound the dissolution rate below 1 nm/h. A direct prior-leverage test separates data leverage from prior leverage, confirming $\mathrm{PBR_{eff}}$ as weakly but genuinely data-determined (imposing the prior this analysis formerly carried moves it by only 4%) and L_0 as prior-set (no interior optimum without it). The identified $\mathrm{PBR_{eff}} = 1.10$ sits a factor of 1.9 below the chromium mass balance for an $FeCr_2O_4$ barrier, and that apparent discrepancy dissolves under examination: the stoichiometric value is phase-dependent and spans $[0.52, 3.79]$ across the candidate barrier spinels, which contains the identified value, and admitting an outer-layer loss term reaches the $FeCr_2O_4$ figure at a χ^2 *decrease* of 0.12. Three exposure times cannot separate oxide production from oxide loss. To our knowledge this is the first formal identifiability characterization of a point-defect-model fit.
4. A physics-informed inverse failure mode has been documented in the corrosion-modeling setting: joint field-and-parameter inversion satisfying the data loss while leaving the governing equation unsatisfied. A hard-mode construction eliminates it, and a forward-solve self-consistency check is proposed for reporting alongside any physics-informed inverse result. This diagnosis is what the exactly solvable test case buys: the neural solver is not needed to obtain the present results—closed-form and Newton references reproduce them—but grading it against those references is what makes it usable where they do not exist, such as self-consistent field-transport coupling or a mobility closure of unknown functional form.
5. Extended spatially with zero additional free parameters, the identified kinetics reproduce the measured Cr-plateau composition from spinel stoichiometry and match the measured layer widths to within ±22%; a structural argument, confirmed numerically over four orders of magnitude in diffusivity, shows quasi-steady composition measurements cannot constrain the defect diffusivity.
6. Across an operating envelope of 200–285°C and 0.01–8 ppm O_2, and across effective activation energies of 3–50 kJ/mol spanning the full plausible range, the ten-year prediction varies by a factor of 1.19 to 1.56—moderately, not negligibly, sensitive to operating condition, with the sensitivity growing with the assumed activation energy.
7. A designed exposure of roughly 3000 h (2σ against the full error budget of future measurement scatter, mechanistic parametric uncertainty and power-law fit uncertainty; roughly 5000–6000 h for 3σ) would discriminate the mechanistic prediction from the power-law extrapolation and would simultaneously improve the identifiability of every weakly constrained parameter.

Data availability

All code implementing the deterministic fits, physics-informed forward and inverse solves, identifiability analyses, and spatial extension is released with PhysicsNeMo's spectral-element-network module (`experimental.models.scen`), together with the example directory containing this paper's model, configuration files, and analysis scripts; exact reproduction commands for every figure and table are documented alongside each script. The digitized Veile et al. layer-thickness data (Table 1) are transcriptions of published figures, released with provenance notes tracing each value to its source figure; this constitutes reuse of published data requiring citation, not separate data-sharing permission.

CRediT authorship contribution statement

Conrard Giresse Tetsassi Feugmo: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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